ZEDPATH

Team Kestrel · IntelliCon '26 Buildathon

ZP-DOC-07

UI/UX Design

Design principles, information architecture, design system, screen specifications and accessibility for the ZedPath app

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Document control

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Version	Date	Author	Description of change
0.9	11 Oct 2026	Team Kestrel	First issue, written from the built app (CR-001 to CR-006 and the video-review fixes included); contrast measured from the colour tokens; submitted for approval

Review and approval

This document takes effect when the approver signs below. Until then its status is as shown on the cover.

Role	Name	Signature	Date
Reviewer	Claude (AI coding partner), consistency and traceability review		
Approver	B.M.P Banneka, Product Owner		

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1 Introduction

1.1 Purpose

This document specifies what a ZedPath user sees and how they move through the app: the principles behind the design, the navigation structure, the design system (colour, type, shape, components), each main screen with its states, the shared interaction patterns, the language and content rules, accessibility, and the performance budget for low-end Android phones on 3G. It records the design **as built** on 11 October 2026, so a reviewer can check every statement against the code.

1.2 Scope

The student-facing web app (a Vite + React single-page app installed as a Progressive Web App, also wrapped as an Android app) for Releases R1 and R2. The curator tools and the API are out of scope; they are covered by ZP-DOC-05 and ZP-DOC-06.

1.3 Intended audience

Designers, developers and testers of ZedPath, and anyone assessing the quality of the user experience (for example, buildathon judges).

1.4 Relationship to other documents

Table 1: Position of this document in the ZedPath document set

Document	Relationship
ZP-DOC-02 User Stories	Parent. Personas P1 to P6 and the stories US-101 to US-708 drive the screens
ZP-DOC-03 SRS	Parent. NFR-001 to NFR-006 (performance) and NFR-020 to NFR-025 (usability, language, accessibility) are the targets here
ZP-DOC-04 Use Case Model	Parent. Each main flow maps to a screen sequence in Section 3
ZP-DOC-05 Data Design	Sibling. Revision rows CR-001 to CR-006 record the data side of the same changes
ZP-DOC-06 Software Architecture	Sibling. Decides the SPA-on-static-assets delivery that the performance budget relies on
ZP-DOC-08 Test Plan	Child. Tests the accessibility checklist and the performance budget
`src/styles.css`, `src/App.tsx`, `src/screens/*.tsx`	Implementation. The design tokens and screens described here
Concept screens (Gate 1)	Reference: https://claude.ai/artifact/N8wGu87fiY8ksUbb82yU8f

1.5 Conventions

Colour and size values are the CSS custom properties ("tokens") in `src/styles.css`; components use tokens only, never raw colours. Routes are written as hash routes (`#/paths`). "Light" and "dark" refer to the theme the phone's system setting selects.

2 Design principles

Table 2: Design principles and how the design applies them

#	Principle	Source	Applied as
DP-1	**Every number links to its source**	Trust rules (CLAUDE.md), US-204, US-602	Cut-offs, rules and answers carry a source line or source chip that opens the UGC handbook page or the official notice
DP-2	**Never "you will get in"**	Trust rules, US-603	Chances are bands (Safe, Likely, Reach) with an explanation, never a promise; Ask gives a verdict chip such as "It depends", never a guarantee
DP-3	**Sample data is labelled as sample**	Trust rules, CR-005	Sample group conversations carry a "Sample" label and cannot be reported or replied to as real; demo mode shows a label above the form
DP-4	**No paid placement**	Trust rules	Order of courses and routes comes only from the student's results and official data
DP-5	**Built for low-end Android on 3G**	NFR-001, NFR-006, interviews E1 to E6	Static SPA, cached app shell, skeletons instead of spinners, one question per screen, no heavy images
DP-6	**Trilingual from the first screen**	NFR-023, US-105	Sinhala, Tamil and English chosen on the Welcome screen; official names and quotes stay in English as published, and the app says so
DP-7	**Safe for minors**	CR-003, CR-005, CR-006	Preset avatars (no photo upload), group rules accepted before taking part, phone numbers and emails refused in messages, report and block
DP-8	**Nothing stored without asking**	Trust rules, US-109	Each contact field states its purpose; consent is recorded; results-based checks for notices run on the phone
DP-9	**One task per screen; validate after interaction**	Material Design 3; ui-ux-pro-max guidance	About you asks one question per screen; errors appear only after Next

3 Personas and key journeys

The personas are defined in ZP-DOC-02. The app is designed first for P1; the others shape specific screens.

Table 3: Personas and the screens designed for them

Persona	Need in one line	Screens designed for them
P1 Kavindu, student with results	"What can I get into, and what do I do next?"	Onboarding, Paths, Courses, Course detail, My list, Journey, Ask
P2 Nimesha, before the exam	Estimate chances before results	About you (estimated Z-score), Paths
P3 Mr. Bandara, parent	Understand the options with the student	Compare, course detail sources, language choice
P4 Ms. Perera, teacher or counsellor	Guide many students consistently	Sourced course pages, Ask with citations
P5 Sahan, verified senior	Answer questions from juniors	Groups (senior badge; seniors land on Groups after login)
P6 Curator	Keep data correct	Out of scope (not in the student app)

Table 4: Key journeys

Journey	Steps (screens)	Stories
J1 First run	Welcome (language) → Intro (3 slides) → Sign up → Save recovery code → About you (5 steps) → Paths	US-105, US-106, US-107, US-109, US-101
J2 Find my chances	Paths (band ring) → Courses (Safe / Likely / Reach tabs) → Course detail	US-201, US-202, US-203, US-204, US-209
J3 Decide	Course detail or Courses (press and hold) → Compare (up to 3) → Add to My list	US-207, US-305, US-301
J4 Build the application list	My list → reorder → order warning → Fix the order → Copy Uni-Codes	US-301, US-302, US-303
J5 Stay on track	Journey: My steps (tick within window) and Open now (notices) → reminders	US-401, US-402, US-404, US-407, US-409 to US-411, US-503
J6 Ask	Ask → suggestion chip or own question → answer card → action (open, add, compare, add to Journey)	US-601 to US-607
J7 Don't decide alone	Groups → accept rules once → room → group info → exit	US-704, US-706, US-707, US-708

4 Information architecture and navigation

4.1 Structure

The app has a **focused account flow** without the navigation bar (Welcome, Intro, Sign up, Log in, Recover, Recovery code, Complete your account, About you) and a **main app** with a top app bar and a five-tab bottom navigation bar. Me opens from the avatar in the top app bar. Routing uses the URL hash so the phone's Back button and deep links work.

Table 5: Bottom navigation tabs

Tab	Icon	Route	Badge
Paths	Map	`#/paths`	none
My list	Ordered list	`#/list`	none
Journey	Calendar	`#/journey`	count of open notices the student has not seen and is not ruled out for (can be turned off in Notification settings)
Groups	People	`#/groups`	none
Ask	Sparkles	`#/ask`	none

4.2 Route table

Table 6: Hash routes, screens and purpose

Route	Screen	Tab	Purpose
`#/`	Welcome	none	Choose language; start or log in
`#/intro`	Intro	none	Three animated slides: paths, bands, sources
`#/signup`, `#/login`, `#/recover`	Account	none	Create an account, log in, reset with the recovery code
`#/saved-code`, `#/saved-code/new`	Recovery code	none	Show the one-time recovery code to save
`#/contact`	Complete your account	none	Accounts made before CR-003 add contact details; also edits them
`#/about`	About you	none	Stream, district, Z-score, subjects and grades, beyond grades
`#/paths`	Your paths (home)	Paths	Band ring, see courses, other routes, My list entry
`#/courses/safe /likely /reach`	Courses	Paths	Courses by band, search, press-and-hold multi-select
`#/hidden`	Hidden courses	Paths	Why each excluded course is out of reach
`#/course/<Uni-Code>`	Course detail	Paths	Needs vs you have, cut-off chart, syllabus, actions
`#/compare`	Compare	My list	Up to 3 courses side by side
`#/list`	My list	My list	Ordered preference list for the UGC application
`#/journey`, `#/journey/open`, `#/alerts`	Journey	Journey	My steps / Open now tabs
`#/routes/<group>`, `#/route/<id>`	Other routes	Paths	Private degrees, job exams, vocational, abroad
`#/groups`	Groups	Groups	Chat list: my groups, popular groups

Route	Screen	Tab	Purpose
`#/group/<id>`, `#/group/<id>/info`	Group room, Group info	Groups	WhatsApp-style room; description, rules, members, exit
`#/ask`, `#/ask/<question>`	Ask ZedPath	Ask	Cited answers; a deep link pre-fills a question
`#/me`	Me	none	Avatar, profile, language, notification settings, account, delete

4.3 Guards

Table 7: Navigation guards

Condition	Result
Not onboarded	Welcome (or Intro) only
Onboarded but no account	Sign up
Account without contact details (pre CR-003)	Complete your account, before anything else
No profile (student)	About you
Senior without a student profile	Groups and Me only

5 Design system

5.1 Colour roles

The palette pairs the Gate 1 deck navy with the logo's amber. Dark mode follows the system setting and redefines every token. Gradients ("aurora") are used for hero moments only: the Welcome screen and the Paths summary card.

Table 8: Colour roles (light theme; dark theme redefines each)

Role	Token	Light	Use
Primary	`--brand` / `--brand-500`	#24478C / #3461C9	Filled buttons (gradient), links, selected states
On primary	`--on-brand`	#FFFFFF	Text on filled buttons
Accent	`--amber`	#FFC94A	Logo, hero highlight
Background	`--bg`	#F4F6FB	Page and app bar (solid)
Surface	`--surface`, `-2`, `-3`	#FFFFFF, #EEF2F9, #E4EAF5	Cards, inputs, tonal buttons
Text	`--ink`, `-2`, `-3`	#0E1525, #4A5468, #7A8396	Body, secondary, helper
Safe	`--safe` on `--safe-bg`	#0F9D63 on #DDF7EA	Safe band
Likely	`--likely` on `--likely-bg`	#2C63E8 on #E3EBFF	Likely band; selected rows
Reach	`--reach` on `--reach-bg`	#B86E00 on #FFF0D2	Reach band
Out of range	`--out` on `--out-bg`	#C9303F on #FDE4E7	Hidden or out-of-range
Not enough data	`--none` on `--none-bg`	#5B6476 on #ECEFF5	No cut-off history
Error	`--danger`	#C9303F	Error text

5.2 Typography

Headings use **Outfit**, body text **Inter**; **Noto Sans Sinhala** and **Noto Sans Tamil** are in both font stacks, so Sinhala and Tamil render with matching weight (NFR-024). Fonts load from Google Fonts with `preconnect`; the system font is the fallback. Numbers use tabular figures; digits stay Western in all three languages, as on results sheets.

Table 9: Type scale

Style	Font	Size / line height	Weight	Use
Display	Outfit	44 / 48	700	Welcome
Screen title	Outfit	28 / 34	600	Screen headings
Title L	Outfit	20 / 26	600	Card titles
App bar title	Outfit	19 / 24	600	Top app bar
Lead	Inter	20 / 28	400	Intro text
Body	Inter	16 / 24	400	Default
Body M	Inter	14 / 21	400	Dense lists
Section label	Inter	12 / 16, uppercase, tracked	600	Section headers
Helper / source	Inter	12 to 13 / 17 to 18	400	Hints, source lines

5.3 Spacing, shape and elevation

Table 10: Layout, shape and elevation tokens

Aspect	Value
Content width	max 640 px, centred; 16 px side gutter; works from 360 px (NFR-025)
Spacing	8 px base: gaps of 8, 10, 14, 16 px; 110 px bottom padding under the navigation bar
Corner radius	`--r-s` 10, `--r-m` 16, `--r-l` 22 (cards), `--r-xl` 28 (hero art); buttons and chips fully rounded
Elevation	`--shadow-1` (cards), `--shadow-2` (raised cards, floating bars), `--shadow-brand` (filled buttons)
Motion	`cubic-bezier(.2,.8,.2,1)`; press feedback scale .97; tab indicator spring; all disabled under reduced motion
Safe areas	`env(safe-area-inset-*)` on Welcome, navigation bar, composer and selection bar

5.4 Components

Table 11: Component catalogue

Component	Specification	Where
Top app bar	Sticky, 60 px, **solid** `--bg` (scrolled text never shows through the title); Back or logo, title, avatar link to Me	All main screens
Navigation bar	Floating, fixed 12 px from the edges above the safe area; 5 equal items (icon 22 px + label); animated pill on the selected item; `aria-current="page"`; count badge on Journey	Main app
Card	`--surface`, radius 22, shadow-1, 16 px padding; `raised` and `aurora` variants	Everywhere
Band pill	Pill with text label (Safe, Likely, Reach, Out of range, Not enough data) in band colours; never colour alone	Courses, detail, Ask, My list
Band ring	116 px donut of Safe / Likely / Reach counts with a legend of numbers	Paths
Buttons	Filled (gradient, 48 px), tonal, white, text (44 px), block (56 px full width); pressed scale .97; disabled 40% opacity	Everywhere
Chips	40 px pill, outline; selected = `aria-pressed`/`aria-checked` filled brand; small chips 34 px	Subjects, interests, Ask suggestions, source chips
Text fields	Label above (`.field`), helper below, error text after interaction	Account, About you
Lists	`list-card` with rows; My list rows have position number, main text, drag grip (44 × 52) and remove	Courses, My list, routes
Selection bar	Sticky above the navigation bar: count, Add to my list / Compare / Remove, Cancel	Courses, My list
Composer	Sticky above the navigation bar: rounded input (48 px) and round 48 px send button	Ask, Group room
Chat bubbles	Question right (brand gradient), answer left; group messages with avatar, name, senior badge	Ask, Groups

Component	Specification	Where
Avatars	12 preset silhouettes in different colours, or initials on a steady colour; decorative (`aria-hidden`)	Top bar, Me, Groups
Skeleton	Grey placeholder blocks while loading	Lists, Compare, Groups, Journey
Alert	`info`, `warn`, `error`, `gold` boxes with `role="status"` or `role="alert"`	Order warnings, errors

6 Screen specifications

States use these meanings. **Loading**: data not yet arrived. **Empty**: nothing to show yet. **Error**: a request failed. **Offline**: the phone has no connection; the cached app shell still opens (service worker), API calls are never served from cache.

6.1 Welcome, Intro and About you (onboarding)

Table 12: Welcome, Intro and About you (onboarding): specification

Aspect	Specification
Purpose	Choose a language, understand what ZedPath does, give results once (US-101, US-105)
Content	Welcome: aurora background, logo, lead line, three language tiles (සිංහල, தமிழ், English), Start and Log in, team line. Intro: three animated slides (paths, Safe/Likely/Reach pills, a sourced number) with dots. About you: five steps, one per screen: stream, district, Z-score, subjects and grades (searchable), beyond grades (achievements and interests)
States	Error: shown under the field only after Next (`role="alert"`). Demo mode: a label above the form says the data is a sample
Interactions	Language applies at once to the whole app; steps slide left or right; Back keeps answers

6.2 Account (sign up, log in, recover, recovery code, complete your account)

Table 13: Account (sign up, log in, recover, recovery code, complete your account): specification

Aspect	Specification
Purpose	A compulsory account with contact details and consent (US-106, US-107, US-109; CR-003)
Content	Username, password; full name, email, Sri Lankan mobile, date of birth, each with its purpose; consent box. Recovery code shown once with copy and "I saved it"
States	Error messages from the server (`role="alert"`); offline is never treated as "no account"
Interactions	After sign-up the student goes straight to About you; a senior goes to Groups

6.3 Paths (home)

Table 14: Paths (home): specification

Aspect	Specification
Purpose	One view of every path the results reach (US-201, US-203, US-501)
Content	Aurora summary card with the band ring and "See courses"; hidden-courses bar with count and "See why"; My list entry; other-route tiles (private, job exams, vocational, abroad) with counts and "Open now" pills
States	Loading: ring and tiles disabled; Error: message from the API; Empty route group: tile disabled

6.4 Courses

Table 15: Courses: specification

Aspect	Specification
Purpose	Browse courses by band (US-203, US-209, US-305)

Aspect	Specification
Content	Tabs Safe / Likely / Reach (`role="tab"`); search across all bands (names, short forms, Uni-Codes); rows with course, institution and band pill; Compare link
States	Loading skeleton; empty search result message
Interactions	Tap opens the course; **press and hold** starts selection; selection bar: Add to my list, Compare (first 3)

6.5 Course detail

Table 16: Course detail: specification

Aspect	Specification
Purpose	Everything about one course, sourced (US-204, US-206, US-208)
Content	Band pill and explanation; **needs vs you have** (each entry rule against the student's subjects); cut-off chart for the student's district with the value on every point; intake, duration, aptitude test; "What you'll study" syllabus by year with its source page; source lines
Interactions	Add to My list or "On your list (#n)"; Compare; Follow (adds the course to the Journey)

6.6 Compare

Table 17: Compare: specification

Aspect	Specification
Purpose	Up to three courses side by side (US-207)
Content	Columns per course: chance, cut-off history, entry rules, duration, places, selection basis, aptitude test; "+" column to add another
States	Loading skeleton per column; nothing estimated here, every value comes from the course API
Interactions	Columns swipe sideways (scroll snap per course); "+" opens a picker (search your courses or pick from My list)

6.7 My list

Table 18: My list: specification

Aspect	Specification
Purpose	The ordered preference list for the UGC form (US-301 to US-303, US-305)
Content	Numbered rows with band pills; how many eligible courses are still left out; order warnings (BR-041, BR-042); Fix the order; Copy Uni-Codes
States	Empty: invitation to add courses from Paths; warnings use `role="status"`
Interactions	Drag by the grip, or focus the grip and use Arrow Up / Down; Fix the order previews and applies only after confirmation; press and hold or "Select" for selection, then **Compare in the order chosen** (max 3) or Remove

6.8 Journey

Table 19: Journey: specification

Aspect	Specification
Purpose	The student's own checklist and dated deadlines (US-401 to US-411, US-503)
Content	Tabs **My steps** and **Open now**. My steps: progress bar, steps (results, list, compare, aptitude tests, deadlines, apply to the UGC with its handbook quote), followed courses and notices. Open now: official notices with closing dates, each showing whether the student meets the A/L and age rules (checked on the phone)
States	Loading skeleton; **locked step**: shows the official rule instead of a date while the UGC has not announced it; Empty: how to follow courses and notices
Interactions	Tick a step only inside its window (server-checked); "I applied" after a notice opens; reminders as phone notifications (Android app) or a calendar file

6.9 Ask ZedPath

Table 20: Ask ZedPath: specification

Aspect	Specification
Purpose	Cited answers in the student's language (US-601 to US-607)
Content	Five **personalised suggestion chips** (chances, best achievement against the reach course, interests, common questions; `src/ask-suggest.ts`); answer card: **verdict chip** (Yes / No / It depends / Not in official sources), lead and bullets with "Show more", band pills checked against the student's results, next-step actions, **source chips** with page numbers, two follow-ups, Helpful / Report a wrong answer
States	Thinking: `role="status"`; Error: `role="alert"` with retry; AI unavailable: core browsing still works (NFR-040)
Interactions	Composer pinned above the navigation bar; question sent with the interface language

6.10 Groups, Group room and Group info

Table 21: Groups, Group room and Group info: specification

Aspect	Specification
Purpose	"Don't decide alone": a group per target course and per A/L batch (US-704 to US-708)
Content	WhatsApp-style chat list (my groups, groups for courses on My list, popular groups); room with bubbles, avatars, names, senior badge; labelled sample conversations; group info: description, rules, members, Exit group
States	Loading skeleton; errors `role="alert"`; reconnecting `role="status"` (only after three failed tries is the connection reported lost)
Interactions	**Rules accepted once per account** at the first join (server-enforced); tap the group name for info; report and block; phone numbers and emails refused

6.11 Me

Table 22: Me: specification

Aspect	Specification
Purpose	The student's own settings (US-102, US-108, US-110)
Content	Avatar picker (12 characters, `radiogroup`-style selection), profile summary and edit, language, **notification settings** (7-day / 1-day reminders, 8 am / 6 pm, badge count; defaults = previous behaviour), contact details, log out, delete account
Interactions	Delete asks for confirmation and removes every server row (cascade)

7 Interaction patterns

Table 23: Shared interaction patterns

Pattern	Behaviour	Why
Press and hold to select	About 450 ms hold starts selection, with a short vibration; moving more than 10 px cancels (it was a scroll); the click after a hold is swallowed; the long-press menu is suppressed (`src/hold.ts`). A visible "Select" button offers the same without holding	Familiar from messaging apps; reviewed in the walkthrough video
Selection order kept	My list Compare opens courses in the order they were picked	The student's priority is preserved
Optimistic updates	Journey ticks, follows and "I applied" show at once and are undone with a message if saving fails	Feels instant on 3G
Reconnect	Group rooms ping every 25 s and reconnect on their own (1, 2, 4 ... up to 15 s) (NFR-043)	No "Disconnected" on a flaky network
Locked steps	A step whose official date is not announced is shown locked with the rule; never an invented date	Trust (DP-1)
Validate after interaction	Errors appear after Next or Submit, not while typing	Less anxiety for first-time users
Deep links	`#/ask/<question>`, `#/course/<code>` and notification links open the right screen; Back works	Sharing and reminders
Demo mode	`?demo=kasun` (off by default) prefills sign-up and About you with the sample student Kasun Perera for the recording; labelled as sample	Gate 3 video; real students always see empty forms

8 Content and language

Table 24: Content rules

Rule	Specification
Tone	Plain, calm, second person; short sentences; no jargon; never promises admission ("Likely", not "you will get in")
Languages	English source strings with Sinhala and Tamil dictionaries (`src/i18n`); TypeScript refuses a missing key; a test checks placeholders, bold tags and script. Marked in-app as AI translation pending native review
Official text	Course and university names, handbook quotes and Gazette notices stay in English as published; Sinhala and Tamil screens say so
Numbers	Every cut-off, intake and date shows its source (handbook page or notice); digits stay Western; tabular figures
Sample data	Sample conversations and demo data are labelled "Sample"; sample messages cannot be reported
Errors	Say what happened and what to do next, in the interface language

9 Accessibility checklist (WCAG 2.2 AA)

Contrast was measured from the colour tokens with the WCAG relative-luminance formula on 11 October 2026.

Table 25: Accessibility checklist and status

Check	Criterion	Status
Body text contrast	1.4.3	Pass: ink on background 16.9:1; secondary text 7.6:1; white on primary 8.9:1; links 5.7:1
Helper text contrast (`--ink-3` #7A8396)	1.4.3	**Gap**: 3.8:1 on white, 3.5:1 on background (needs 4.5:1 for 12 to 13 px text). Action: darken to about #646D80. Dark theme passes (5.5:1)
Band pill contrast (light)	1.4.3	**Gap**: Safe 3.1:1, Reach 3.5:1, Likely 4.3:1, Out 4.4:1 at 12 px bold; Not enough data passes (5.2:1). Action: darken the band ink colours. Dark theme passes (6.5 to 8.8:1)
Colour not the only cue	1.4.1	Pass: every band pill carries a text label
Touch targets	2.5.8 (24 px min); NFR-022 (48 dp)	Pass for 2.5.8. Buttons 48 px, icon buttons 44 px, nav items 52 px, grips 44 × 52. **Gap against NFR-022**: chips 40 px, small chips 34 px
Focus visible	2.4.7, 2.4.11	Pass: 3 px outline on `:focus-visible` everywhere
Keyboard	2.1.1	Pass: list reorder by Arrow keys on the grip; "Select" button alternative to press and hold
Names and roles	4.1.2	Icon buttons have `aria-label`; tabs use `role="tab"` and `aria-selected`; navigation uses `aria-current`; chips use `aria-pressed`/`aria-checked`
Status messages	4.1.3	`role="status"` for warnings and progress, `role="alert"` for errors
Decorative images	1.1.1	Avatars, logo and icons are `aria-hidden` (the name is shown beside them)
Reduced motion	2.3.3	All transitions and animations off under `prefers-reduced-motion`
Reflow	1.4.10; NFR-025	Pass: single column, max 640 px, from 360 px without horizontal scroll
Language	3.1.1	`lang` attribute is "en"; **gap**: not updated when Sinhala or Tamil is chosen. Action: set `document.documentElement.lang`

The gaps are logged for the next design iteration; none blocks a flow.

10 Performance budget for 3G

Table 26: Performance budget

Measure	Budget	How the design meets it
First screen interactive (Fast 3G, mid-range Android)	3 s (NFR-001)	Static SPA from Cloudflare static assets (no Worker on page loads, NFR-012); service worker caches the shell and hashed assets
Results shown after About you	2 s (NFR-002)	One `POST /results` call; skeletons while waiting
Initial JavaScript	200 KB compressed (NFR-006)	Single bundle; icons from one tree-shaken set; no images on the critical path except SVGs
API response	500 ms p95 (NFR-003)	ETag revalidation on the API; API responses are never cached by the service worker, so data is always current
Fonts	Text visible at once	`preconnect` to the font host; `display=swap`, so system fonts show until the web fonts arrive
Ask	First answer within 4 s p95 (NFR-004)	"Thinking" status; the rest of the app does not depend on Ask (NFR-040)

11 Usability evidence

The design changed after evidence from users and the product owner's review videos. The data side of each change is recorded in the ZP-DOC-05 revision rows; the story side in ZP-DOC-02.

Table 27: Changes driven by usability evidence

Change	Evidence	What changed in the UI
CR-001	Product owner's change request (5 Oct): the profile should follow the student between devices	Optional account, recovery code screen
CR-002	Product owner's GO (11 Oct), building on the deadline and "ask a senior" needs in ZP-DOC-02	Journey, Alerts, Groups; five-tab navigation
CR-003	Product owner's decision (contact for reminders, consent law)	Compulsory account with four details, each with purpose and consent
CR-004	Review: "AI replies are not formatted well"	Ask answer card: verdict chip, bullets, source chips, actions, feedback
CR-005	Walkthrough video 2 (8 min 50 s): "groups should have rules", "WhatsApp-style groups"	Rules sheet, WhatsApp-style list and room, labelled samples; Ask box pinned to the bottom; cut-off values on chart points; Compare columns swipe with "+"
CR-006	Product owner's decisions of 11 Oct after review	Rules accepted once, group info page, avatars, rooms that reconnect, My list selection with Compare, Journey ticks within window, Alerts moved into Journey as "Open now"
Other review fixes	Same review round	Solid top app bar (scrolled text showed through the title); "Sit the A/L again" removed; press-and-hold multi-select on Courses; test-notification button removed from Me

12 Traceability

Table 28: Traceability from requirements to this document

Requirement or story	Section of this document
NFR-001 to NFR-006, NFR-012	Performance budget for 3G
NFR-020	Personas and key journeys (J1 to J4)
NFR-021, NFR-022, NFR-025	Accessibility checklist
NFR-023, NFR-024, US-105	Typography; Content and language
NFR-040, NFR-043	Ask ZedPath; Interaction patterns
US-101 to US-111	Welcome, Intro and About you; Account; Me
US-201 to US-209	Paths; Courses; Course detail; Compare
US-301 to US-305	My list; Interaction patterns (press and hold)
US-401 to US-411, US-501 to US-503	Journey; Paths (other routes)
US-601 to US-607	Ask ZedPath
US-704 to US-708	Groups, Group room and Group info
Trust rules (CLAUDE.md)	Design principles DP-1 to DP-4; Content and language